

Neural ODE vs. ResNet: An Empirical Comparison on Image Classification

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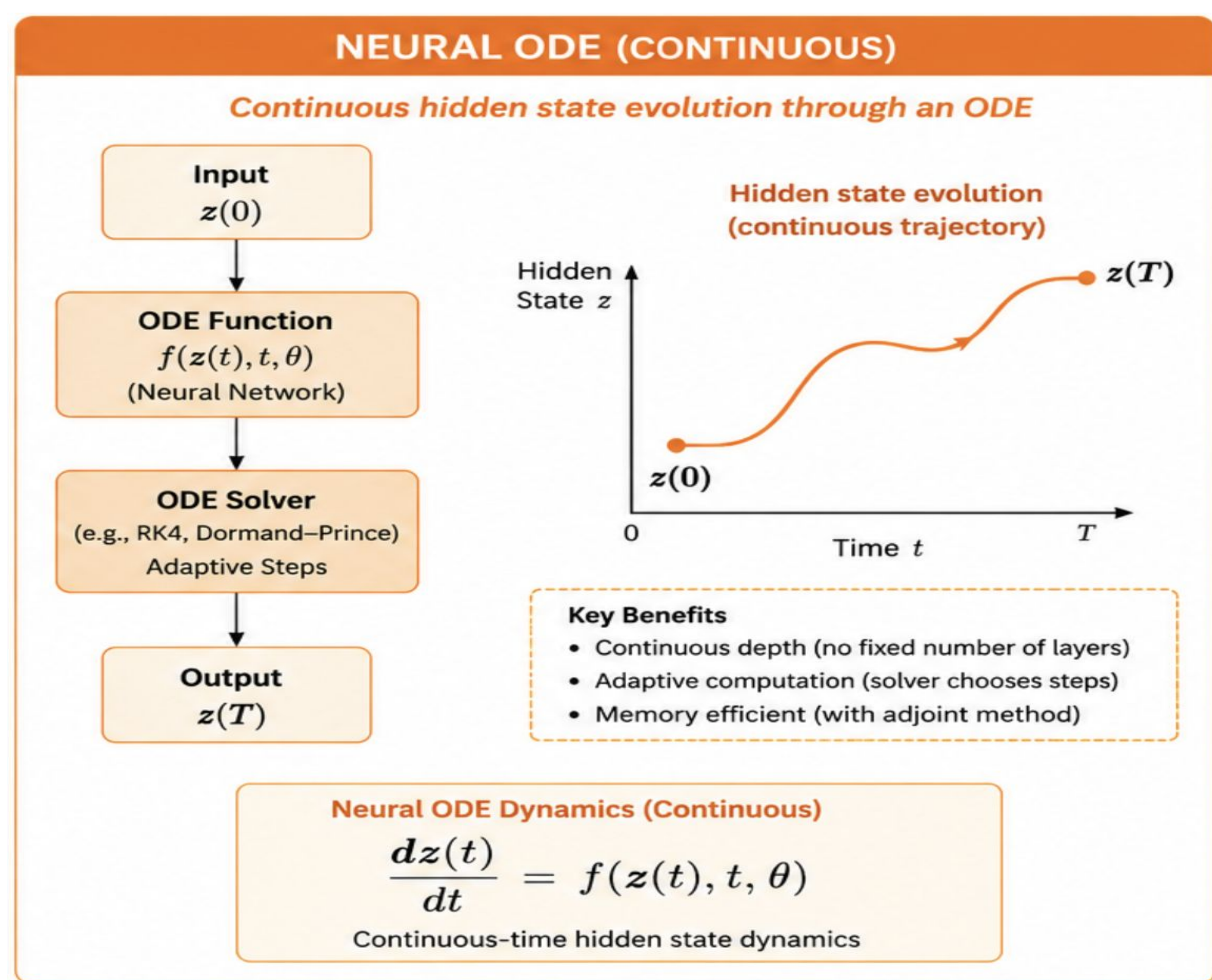
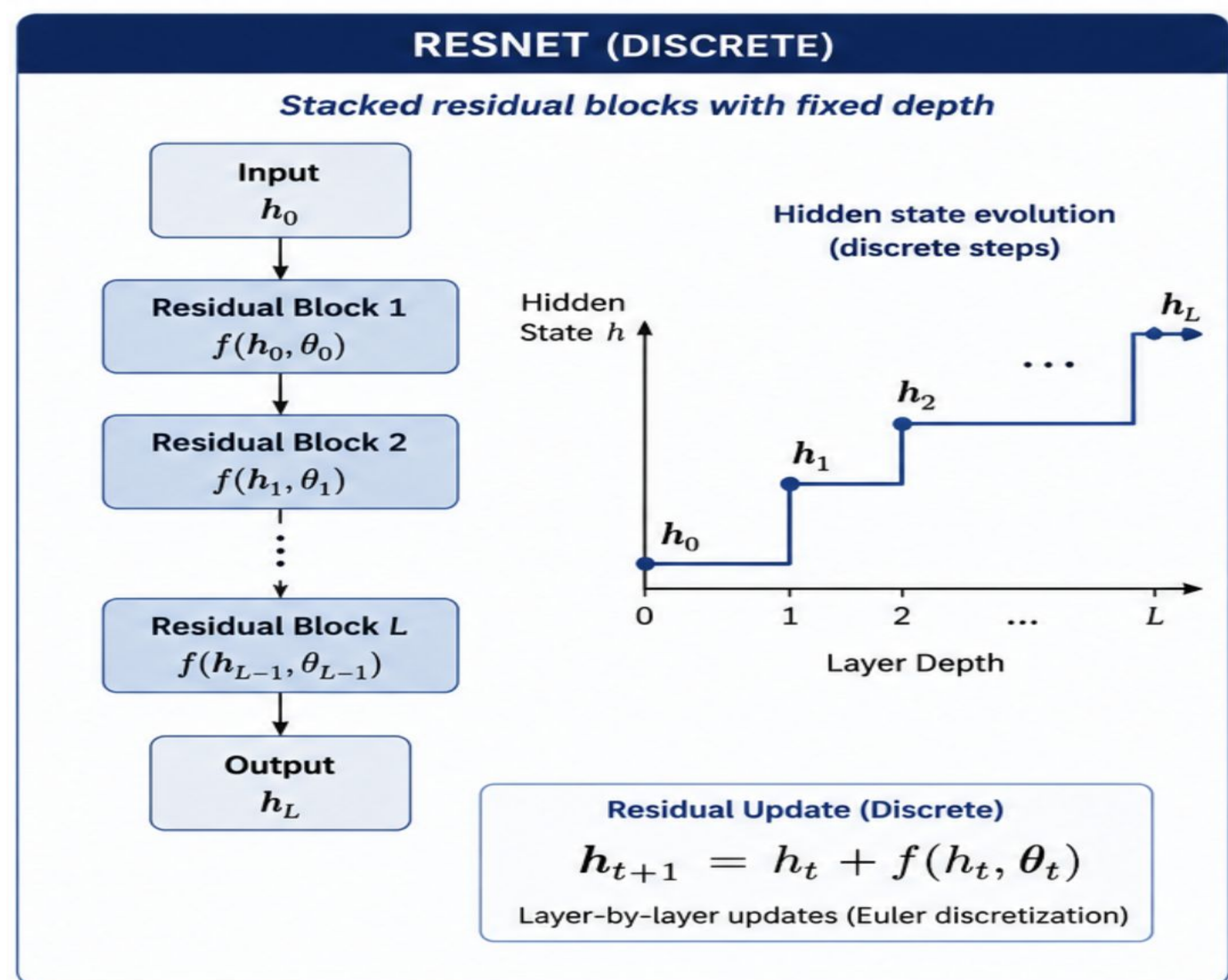
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MOTIVATION

Traditional deep neural networks rely on stacking discrete layers to learn complex representations. Residual Networks (ResNets) improved training stability through residual connections, but still require fixed computational depth and large memory usage. Neural Ordinary Differential Equations (Neural ODEs) reinterpret residual learning as a continuous dynamical system, where hidden states evolve continuously through time using differential equations.

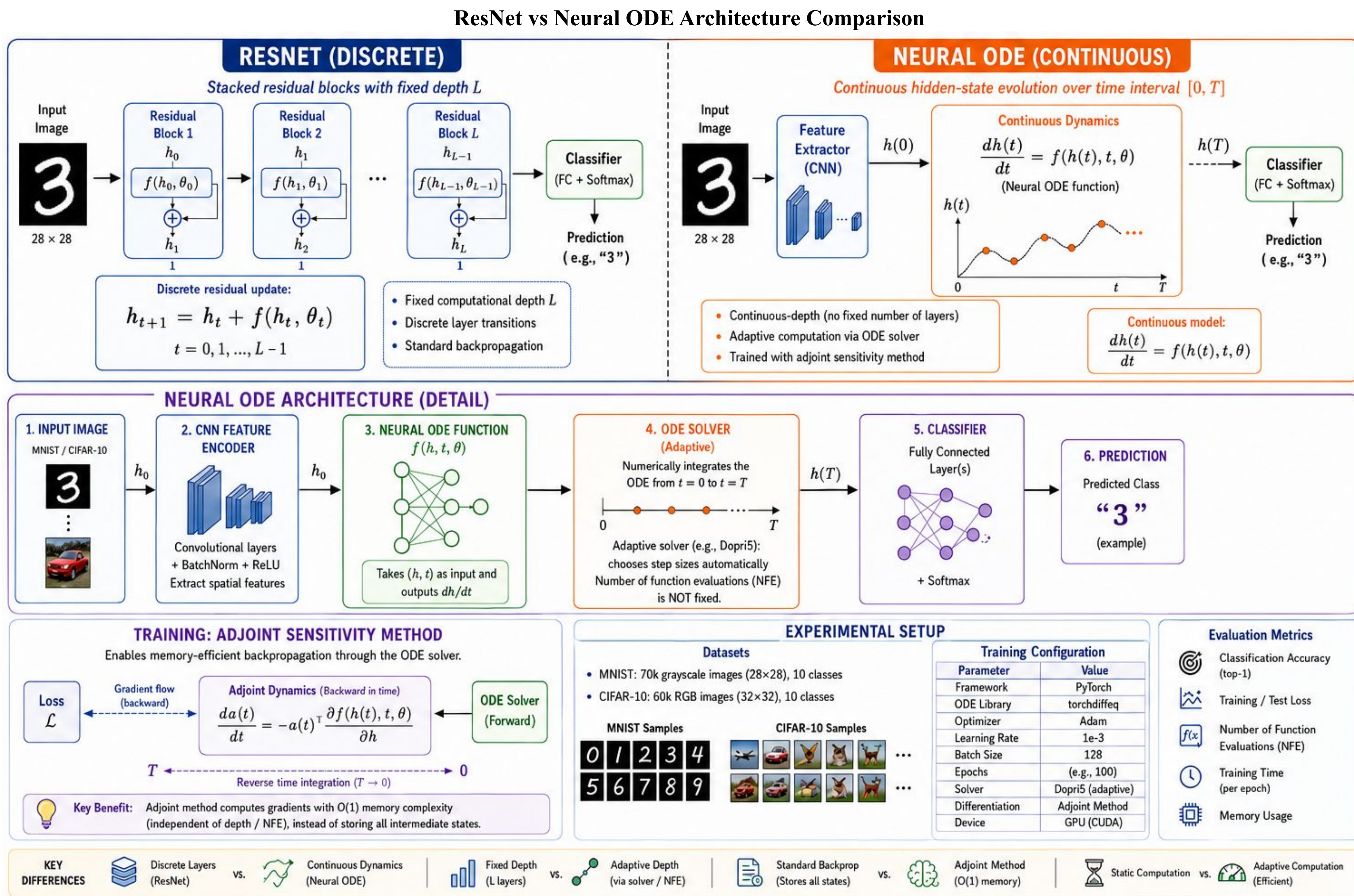
GOAL

Re-implement and compare ResNet and Neural ODE architectures on MNIST and CIFAR-10 image classification tasks.



- ResNets can be interpreted as Euler discretizations of differential equations.
- Neural ODEs replace discrete layers with continuous dynamics.
- ODE solvers adaptively determine computational depth.
- Continuous-depth models provide memory-efficient training.

METHODOLOGY + METRICS



RESULTS + ANALYSIS

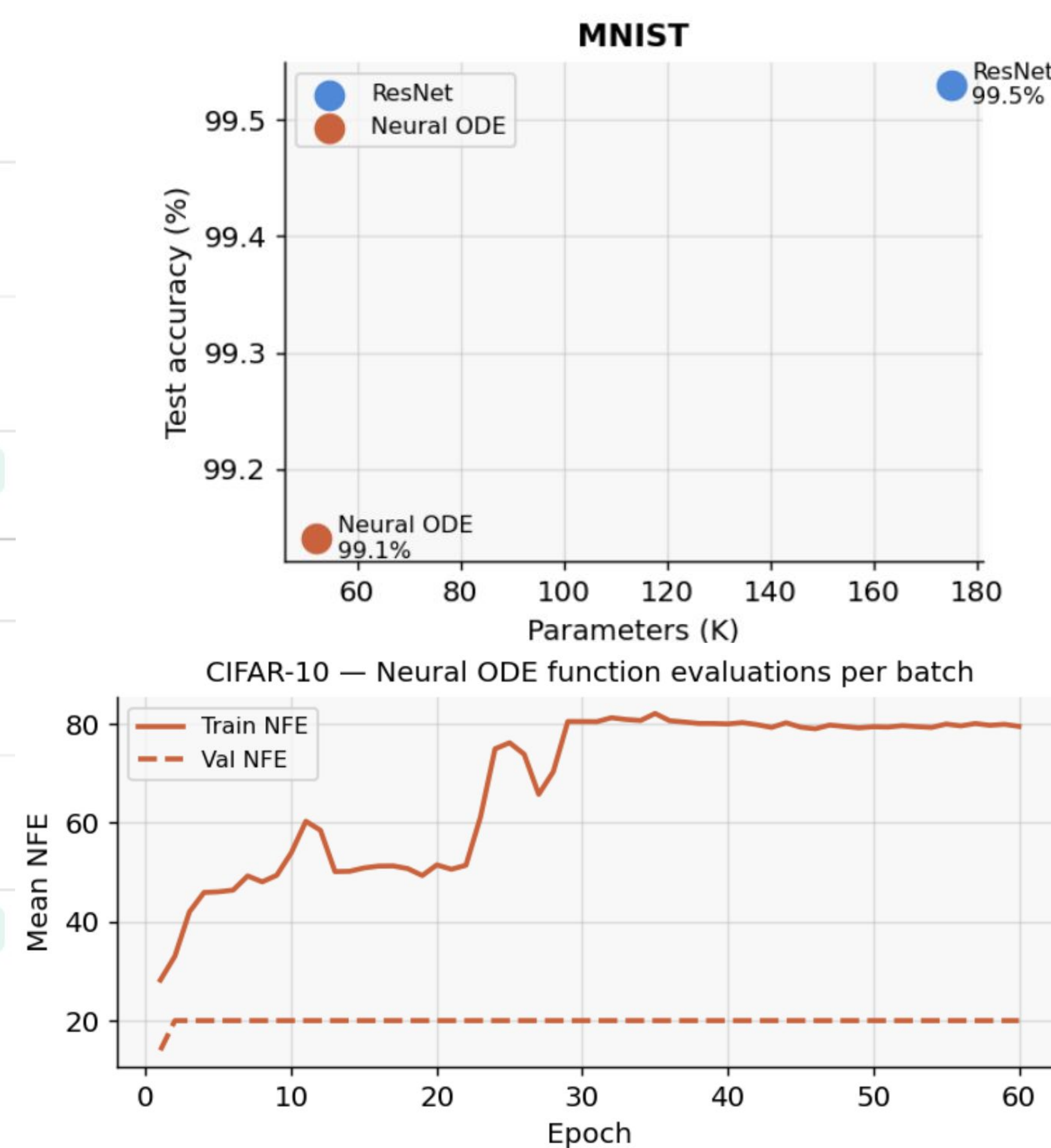
Results				
MNIST	Test acc	Params	Train time	Peak mem
ResNet 174K params	99.53%	174,970	125s	145 MB
Neural ODE 52K params	99.14%	52,138	1,058s	121 MB
		3.4x fewer params		16% less memory
CIFAR-10	Test acc	Params	Train time	Peak mem
ResNet 696K params	90.84%	696,618	553s	519 MB
Neural ODE 207K params	83.99%	207,818	2,612s	289 MB
		3.4x fewer params		44% less memory

Key Findings

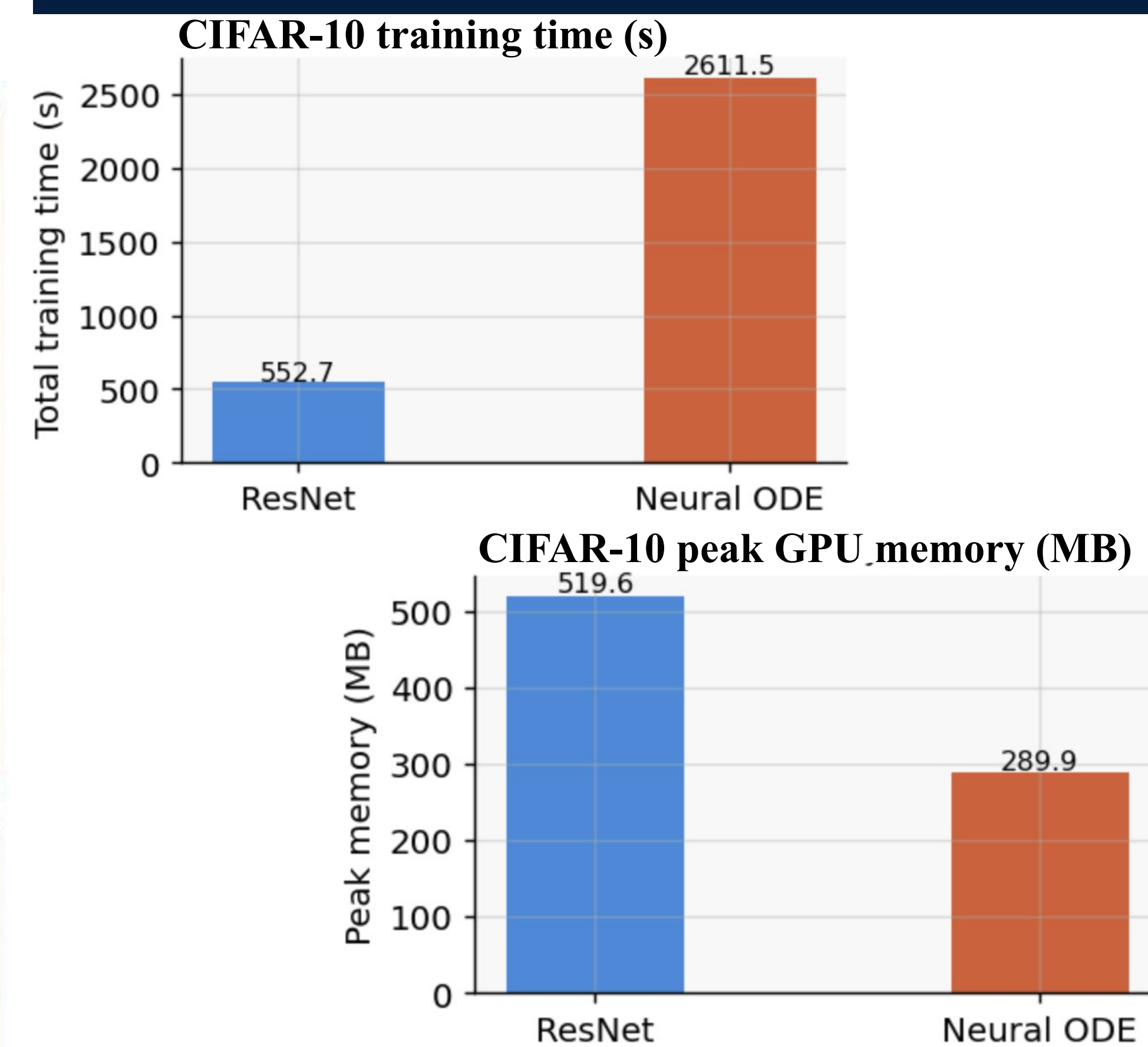
Parameter efficiency
Near-identical MNIST accuracy with 3.4x fewer parameters

Memory advantage
Adjoint method cuts peak GPU memory by up to 44%

Compute tradeoff
NFE ~67/batch $\rightarrow$ 8.4x slower training vs ResNet on MNIST



DISCUSSION AND CONCLUSIONS



Conclusion. Neural ODEs are a strong ResNet alternative when memory and parameter efficiency matter more than training speed. For simple tasks they are competitive; for complex datasets, ODE solving overhead remains a barrier.

FUTURE WORK

- Explore fixed-step solvers (Euler, RK4) as faster alternatives to adaptive *dopri5* and measure the accuracy-speed tradeoff directly.
- Scale Neural ODE to deeper feature extractors to close the CIFAR-10 accuracy gap against ResNet.
- Benchmark on more complex datasets (CIFAR-100, TinyImageNet) to better characterize where continuous-depth models break down.
- Investigate learned integration time intervals instead of fixed $t \in [0, 1]$ to give the ODE more expressive flexibility.
- Explore hybrid architectures combining ResNet stages with a single ODE block to balance accuracy, memory, and training speed.

References

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Acknowledgments

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